PROJECT DEVELOPMENT PHASE

Model Performance Test

Date	10 February 2025
Team ID	SWTID-2026-3075
Team Members	Karpagam R (Leader), Padma PriyaR, Mariyam Asiya S,Shreya S (Team Size: 4)
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders (ServiceNow)
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot / Proof Output
1.	**Metrics** (Model Validation & Evaluation)	**Regression Model (Procurement Turnaround):** • MAE: **0.042** • MSE: **0.0031** • RMSE: **0.0557** • R² Score: **0.964** **Classification Model (Order Routing):** • Confusion Matrix: **[[142, 8], [6, 144]]** • Accuracy Score: **95.33%** • Classification Report: **Precision: 0.95 | Recall: 0.96 | F1: 0.95**	Console > evaluate_model.py [INFO] Executing Evaluation Pipeline... MAE: 0.0420 | RMSE: 0.0557 R2_SCORE: 0.9642 (Optimized) CONFUSION MATRIX: [[142, 8], [6, 144]] ACCURACY: 95.33% | STATUS: OK
2.	**Tune the Model** (Optimization)	**Hyperparameter Tuning:** • Grid Search CV (n_estimators=100, max_depth=12, learning_rate=0.05) • Loss Function: Log-Loss / Huber Loss **Validation Method:** • Stratified 5-Fold Cross Validation (80:20 Train-Test Split with automated feature scaling)	ServiceNow Studio > ML Tuning [TUNING] Method: Stratified 5-Fold Hyperparameters: {max_depth: 12, lr: 0.05} Fold 1: 95.1% | Fold 2: 95.6% | Fold 3: 95.0% Fold 4: 95.4% | Fold 5: 95.7% ✓ Convergence achieved in 48 iterations

FUNCTIONAL & PERFORMANCE TESTING TEMPLATE

Model Performance Test

Date	21 February 2025
Team ID	SWTID-2026-1920
Team Leader & Members	Karpagam R (Leader), Padma PriyaR, Mariyam Asiya S,Shreya S
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders (ServiceNow)
Maximum Marks	10 Marks

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Text Input Validation (e.g., Department, Laptop Tier)	Enter standard laptop catalog inputs and invalid string payloads into fields	Valid inputs accepted, clear validation error flags for improper formats	Verified as expected	**Pass**
FT-02	Number Input Validation (e.g., RAM size, Quantity)	Enter order quantity within (1-10) and boundary/negative test values	Accepts valid counts; throws validation warning for out-of-range bounds	Warning rendered	**Pass**
FT-03	Order / Content Generation (Automated PO & Workflow)	Submit finalized laptop bundle and trigger automated approval flow	Correct ServiceNow RITM generated with auto-assigned approver tasks	Workflow initiated	**Pass**
FT-04	API Connection Check	Trigger ServiceNow IntegrationHub REST call to asset inventory system	API responds with HTTP 200 OK status code and updated inventory count	HTTP 200 OK	**Pass**
PT-01	Response Time Test	Measure end-to-end checkout execution time on catalog order submission	Transaction round-trip latency must complete in under 3.0 seconds	1.42 seconds	**Pass**
PT-02	API Speed / Concurrency Test	Simulate 50 simultaneous laptop requisition requests via automated scripts	API handles concurrent requests without latency spikes or queue drops	Avg 210ms / call	**Pass**
PT-03	File Upload Load Test (e.g., Hardware Specs & Invoices)	Upload multiple hardware specification PDF files simultaneously	Upload completes successfully with virus scan pass and zero memory leaks	Processed 100%	**Pass**